

Linear Algebra

Linear Algebra

Author

MUHO CHOI

Email: sihyeon8240@gmail.com

GitHub: <https://github.com/sihyeon8240>

Source Code & Contributions

GitHub: <https://github.com/sihyeon8240/math>

Feel free to open issues or pull requests for corrections.

© 2026. MUHO CHOI.

This textbook is licensed under a [Creative Commons Attribution 4.0 International License](https://creativecommons.org/licenses/by/4.0/) (CC BY 4.0).

You are free to: Share, copy, and redistribute the material in any medium or format; adapt, remix, transform, and build upon the material for any purpose, even commercially.

Under the following terms: You must give appropriate credit, provide a link to the license, and indicate if changes were made.

Contents

Preface	v
1 Vector Spaces	1
1.1 Definitions	1
1.2 Bases	1
1.3 Dimension of a Vector Space	1
1.4 Sums and Direct Sums	1
2 Matrices	2
2.1 The Space of Matrices	2
2.2 Linear Equations	2
2.3 Multiplication of Matrices	2
3 Linear Mappings	3
3.1 Mappings	3
3.2 Linear Mappings	3
3.3 The Kernel and Image of a Linear Map	3
3.4 Composition and Inverse of Linear Mappings	3
3.5 Geometric Applications	3
4 Linear Maps and Matrices	4
4.1 The Linear Map Associated with a Matrix	4
4.2 The Matrix Associated with a Linear Map	4
4.3 Bases, Matrices, and Linear Maps	4
5 Scalar Products and Orthogonality	5
5.1 Scalar Products	5
5.2 Orthogonal Bases, Positive Definite Case	5
5.3 Application to Linear Equations; the Rank	5
5.4 Bilinear Maps and Matrices	5
5.5 General Orthogonal Bases	5
5.6 The Dual Space and Scalar Products	5
5.7 Quadratic Forms	5

5.8	Sylvester's Theorem	5
6	Determinants	6
6.1	Determinants of Order 2	6
6.2	Existence of Determinants	6
6.3	Additional Properties of Determinants	6
6.4	Cramer's Rule	6
6.5	Triangulation of a Matrix by Column Operations	6
6.6	Permutations	6
6.7	Expansion Formula and Uniqueness of Determinants	6
6.8	Inverse of a Matrix	6
6.9	The Rank of a Matrix and Subdeterminants	6
7	Symmetric, Hermitian, and Unitary Operators	7
7.1	Symmetric Operators	7
7.2	Hermitian Operators	7
7.3	Unitary Operators	7
8	Eigenvectors and Eigenvalues	8
8.1	Eigenvectors and Eigenvalues	8
8.2	The Characteristic Polynomial	8
8.3	Eigenvalues and Eigenvectors of Symmetric Matrices	8
8.4	Diagonalizations of a Symmetric Linear Map	8
8.5	The Hermitian Case	8
8.6	Unitary Operators	8
9	Polynomials and Matrices	9
9.1	Polynomials	9
9.2	Polynomials of Matrices and Linear Maps	9
10	Triangulation of Matrices and Linear Maps	10
10.1	Existence of Triangulation	10
10.2	Theorem of Hamilton-Cayley	10
10.3	Diagonalization of Unitary Maps	10
11	Polynomials and Primary Decomposition	11
11.1	The Euclidean Algorithm	11
11.2	Greatest Common Divisor	11
11.3	Unique Factorization	11
11.4	Application to the Decomposition of a Vector Space	11
11.5	Schur's Lemma	11
11.6	The Jordan Normal Form	11

12 Convex Sets	12
12.1 Definitions	12
12.2 Separating Hyperplanes	12
12.3 Extreme Points and Supporting Hyperplanes	12
12.4 The Krein-Milman Theorem	12

Preface

This textbook uses Lean 4 to formalize selected mathematical statements and proofs.

- **Tools.** Lean is an interactive theorem prover; Mathlib is a library of formalized mathematics. This edition uses Lean 4.33.1 and Mathlib v4.33.1 (revision [0df444a360ea](#)).
- **Reading the proofs.** Each paired statement or proof step gives a mathematical explanation followed by its Lean code. The explanations can be read independently; code fragments are read in order within the complete declaration.
- **Verification.** Lean checks formal proofs under their stated assumptions. Verification covers only statements linked to checked declarations. Agreement between the explanations, displayed code, and checked proofs is reviewed separately.
- **Prerequisites.** Starting assumptions and accepted external results are specific to this textbook.

Chapter 1

Vector Spaces

1.1 Definitions

1.2 Bases

1.3 Dimension of a Vector Space

1.4 Sums and Direct Sums

Chapter 2

Matrices

2.1 The Space of Matrices

2.2 Linear Equations

2.3 Multiplication of Matrices

Chapter 3

Linear Mappings

3.1 Mappings

3.2 Linear Mappings

3.3 The Kernel and Image of a Linear Map

3.4 Composition and Inverse of Linear Mappings

3.5 Geometric Applications

Chapter 4

Linear Maps and Matrices

4.1 The Linear Map Associated with a Matrix

4.2 The Matrix Associated with a Linear Map

4.3 Bases, Matrices, and Linear Maps

Chapter 5

Scalar Products and Orthogonality

5.1 Scalar Products

5.2 Orthogonal Bases, Positive Definite Case

5.3 Application to Linear Equations; the Rank

5.4 Bilinear Maps and Matrices

5.5 General Orthogonal Bases

5.6 The Dual Space and Scalar Products

5.7 Quadratic Forms

5.8 Sylvester's Theorem

Chapter 6

Determinants

- 6.1 Determinants of Order 2**
- 6.2 Existence of Determinants**
- 6.3 Additional Properties of Determinants**
- 6.4 Cramer's Rule**
- 6.5 Triangulation of a Matrix by Column Operations**
- 6.6 Permutations**
- 6.7 Expansion Formula and Uniqueness of Determinants**
- 6.8 Inverse of a Matrix**
- 6.9 The Rank of a Matrix and Subdeterminants**

Chapter 7

Symmetric, Hermitian, and Unitary Operators

7.1 Symmetric Operators

7.2 Hermitian Operators

7.3 Unitary Operators

Chapter 8

Eigenvectors and Eigenvalues

8.1 Eigenvectors and Eigenvalues

8.2 The Characteristic Polynomial

8.3 Eigenvalues and Eigenvectors of Symmetric Matrices

8.4 Diagonalizations of a Symmetric Linear Map

8.5 The Hermitian Case

8.6 Unitary Operators

Chapter 9

Polynomials and Matrices

9.1 Polynomials

9.2 Polynomials of Matrices and Linear Maps

Chapter 10

Triangulation of Matrices and Linear Maps

10.1 Existence of Triangulation

10.2 Theorem of Hamilton-Cayley

10.3 Diagonalization of Unitary Maps

Chapter 11

Polynomials and Primary Decomposition

11.1 The Euclidean Algorithm

11.2 Greatest Common Divisor

11.3 Unique Factorization

11.4 Application to the Decomposition of a Vector Space

11.5 Schur's Lemma

11.6 The Jordan Normal Form

Chapter 12

Convex Sets

12.1 Definitions

12.2 Separating Hyperplanes

12.3 Extreme Points and Supporting Hyperplanes

12.4 The Krein-Milman Theorem

Bibliography

- [1] Serge Lang, *Linear Algebra*, Springer Nature.